

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Disconnect the batteries. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>).
- Disconnect the air starter to reduce the possibility of accidental starting. Refer to the OEM Service Manual.
- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

- Remove the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>).
- For engines that do **not** have a fan drive, remove the thermostat housing support. Refer to Procedure 008-015. (</qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html>) Remove the LTA water pump. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>).
- Remove the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html>).
- Remove the alternator and bracket. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/56/56-013-001-tr.html>).
- Remove the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html>).
- Remove the alternator and accessory drive pulleys, if equipped. Refer to Procedure 009-004 (</qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html>).
- Remove the vibration dampers. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html>).
- Remove the fan hub idler. Refer to Procedure 008-029 (</qs3/pubsys2/xml/en/procedures/56/56-008-029-tr.html>).
- Remove the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/56/56-001-023-tr.html>).
- Remove the front engine support bracket. Refer to Procedure 016-002 (</qs3/pubsys2/xml/en/procedures/56/56-016-002-tr.html>).

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

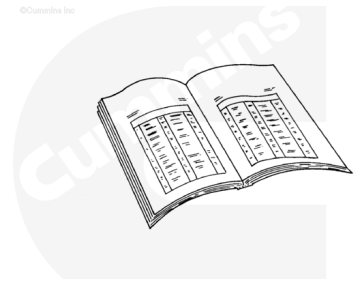
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Disconnect the batteries. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>).
- Disconnect the air starter to reduce the possibility of accidental starting. Refer to the OEM Service Manual.
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>).
- For engines that do **not** have a fan drive, remove the thermostat housing support. Refer to Procedure 008-015 (</qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html>).
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>).
- Remove the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html>).
- Remove the alternator and bracket. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/56/56-013-001-tr.html>).
- Remove the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html>).



ck800wa

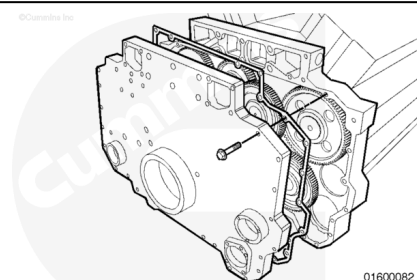
- Remove the alternator and accessory drive pulleys, if equipped. Refer to Procedure 009-004 (</qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html>).
- Remove the vibration dampers. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html>).
- Remove the fan hub idler. Refer to Procedure 008-029 (</qs3/pubsys2/xml/en/procedures/56/56-008-029-tr.html>).
- Remove the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/56/56-001-023-tr.html>).
- Remove the front engine support bracket. Refer to Procedure 016-002 (</qs3/pubsys2/xml/en/procedures/56/56-016-002-tr.html>).

Remove

with Mechanically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



The front cover has capscrews that go up through the oil pan adapter and into the front gear cover.

Remove the 28 capscrews and washers.

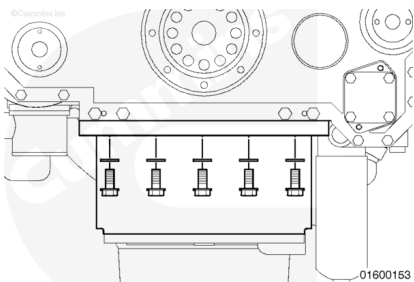
Use the guide pins to support the front gear cover, slide it forward, and install the lifting eye into the threaded hole in the top center of the front gear cover.

Remove the front gear cover and gear cover gasket.

The front gear cover gasket is an edge-molded gasket. Inspect the gasket; if any damage is found, the gasket **must** be replaced.

with Electronically Actuated Injector

Remove the five capscrews that go up through the oil pan adapter and into the front gear cover.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

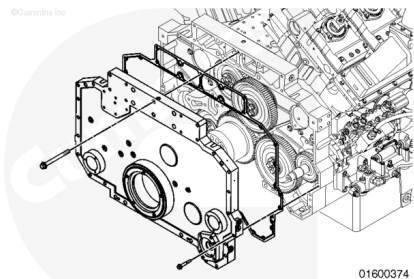
Remove the front gear cover capscrews and washers.

Note : The number of capscrews depends on the engine options.

Use the guide pins to support the front gear cover, slide it forward, and install the lifting eye into the threaded hole in the top center of the front gear cover.

Remove the front gear cover and gear cover gasket.

The front gear cover gasket is an edge-molded gasket. Inspect the gasket; if any damage is found, the gasket **must** be replaced.

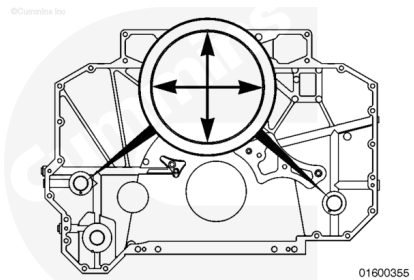


Inspect for Reuse

with Mechanically Actuated Injector

Measure the inside diameter of the water pump and accessory drive bushings.

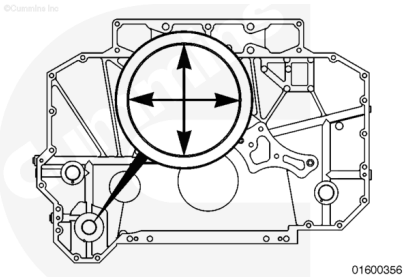
mm		in
43.99	MIN	1.732
44.07	MAX	1.735



If the bushings are **not** within specifications. The bushings **must** be replaced.

Measure the inside diameter of the hydraulic pump drive bushing inside diameter.

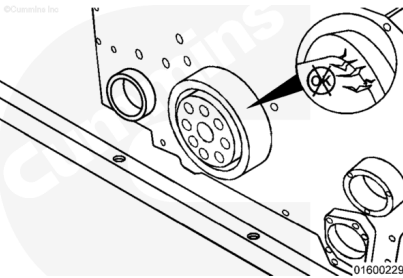
mm		in
43.99	MIN	1.732
44.07	MAX	1.735



If the bushing is **not** within specification. The bushing **must** be replaced.

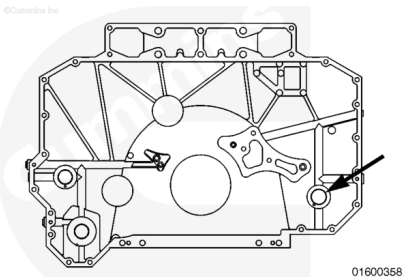
Check the gear cover for cracks or damage. If the gear cover is damaged, it **must** be replaced.

Use emery paper to remove any nicks or burrs from the outside diameter of the trunnion.



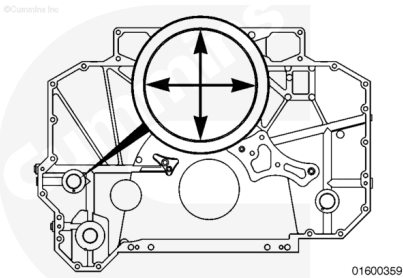
with Electronically Actuated Injector

There is no gear cover bushing for the water pump on engine equipped with electronically actuated injectors. A sleeve is installed in this location to block the oil passage.



Measure the inside diameter of the fuel pump drive bushings.

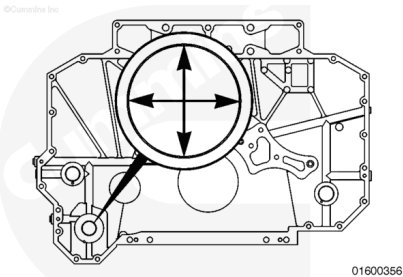
mm		in
43.99	MIN	1.732
44.07	MAX	1.735



If the bushing is **not** within specifications. The bushing **must** be replaced.

Measure the inside diameter of the hydraulic pump drive bushing inside diameter.

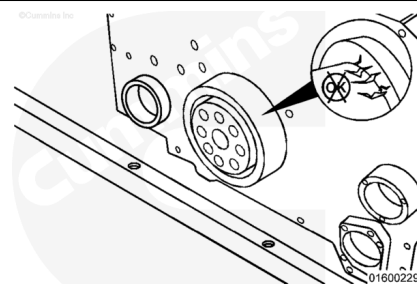
mm		in
43.99	MIN	1.732
44.07	MAX	1.735



If the bushing is **not** within specification. The bushing **must** be replaced.

Check the gear cover for cracks or damage. If the gear cover is damaged, it **must** be replaced.

Use emery paper to remove any nicks or burrs from the outside diameter of the trunnion.



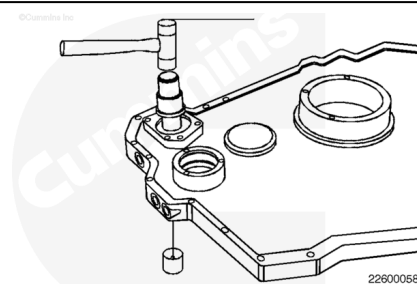
Disassemble

The bushings can **only** be removed from the front gear cover after the gear cover has been removed.

The bushings can be removed from either the front or rear of the front gear cover.

Place the bushing remover against the bushing, and strike the tool with a hammer or a mallet until the bushing is removed.

Repeat the step for the remaining bushings.

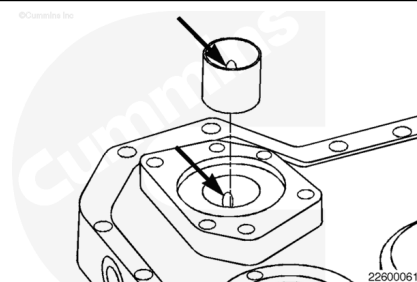


Assemble

⚠ CAUTION ⚠

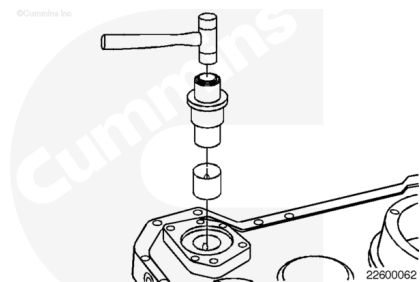
Failure to align the bushing oil hole with the hole in the gear cover will prevent proper lubrication of the bushing.

Place the bushing into the bore and align the bushing oil hole with the oil hole inside of the bore.



Check the alignment of the bushing and the gear cover oil holes with a drill bit or a small piece of rod. If the holes are **not** aligned, remove the bushing and align the holes.

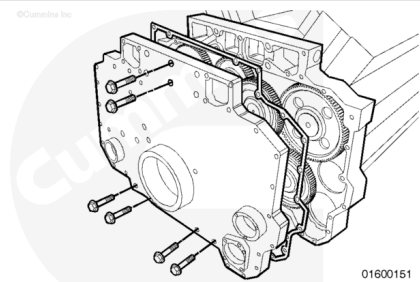
Strike the installation tool firmly with a mallet or a hammer. Continue to strike the tool until the bushing is installed. Repeat the procedure for remaining bushings.



Install

with Mechanically Actuated Injector

This illustration shows the location for the six 16 mm capscrews. All of the other front cover capscrews are 12 mm.



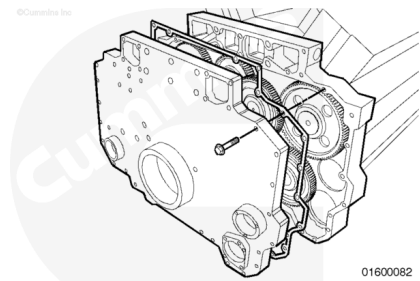
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Do **not** tighten the capscrews until the master dowel pin is installed.

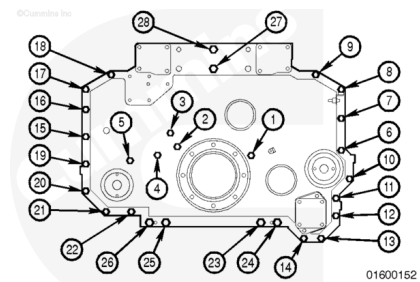
The use of guide bolts will aid the assembly.

Install the front cover gasket, cover, and capscrews.



Tighten the 28 mounting capscrews in the sequence shown.

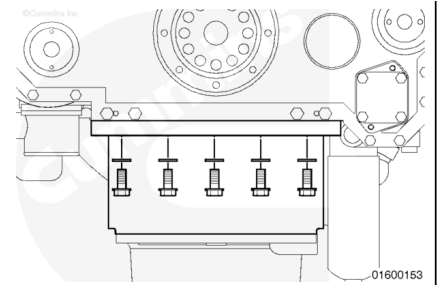
M12 capscrews	80 n.m	[59 ft-lb]
M16 capscrews	195 n.m	[144 ft-lb]



Install the five capscrews that attach to the bottom of the oil pan adapter.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



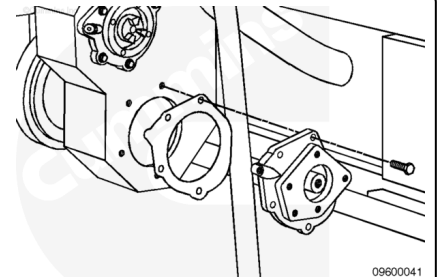
Install the hydraulic pump drive.

Use clean engine oil to lubricate the bushing in the front gear cover.

Install the gasket, drive or cover, washers, and capscrews.

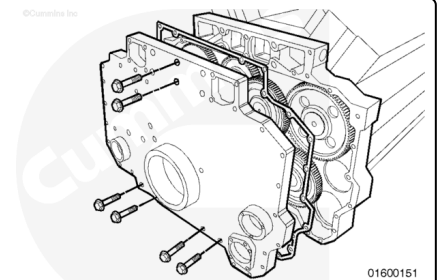
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



with Electronically Actuated Injector

This illustration shows the location for the six 16 mm capscrews. All of the other front cover capscrews are 12 mm.



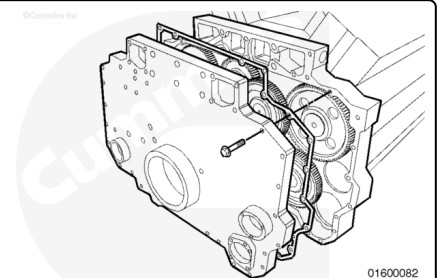
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Do **not** tighten the capscrews until the master dowel pin is installed.

The use of guide bolts will aid the assembly.

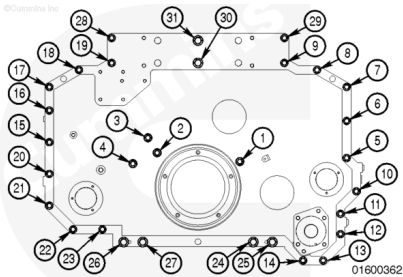
Install the front cover gasket, cover, and capscrews.



Note : The number of capscrews that can be tightened at this time depends on the engine options. If the location is used to mount an accessory, it can **not** be installed at this time.

Tighten the front gear cover mounting capscrews in the sequence shown.

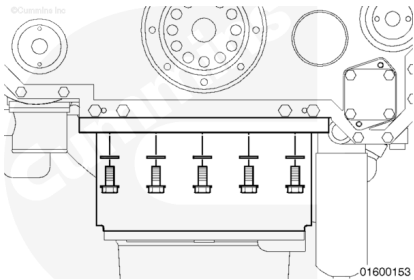
M12 capscrews	80 n.m	[59 ft-lb]
M16 capscrews	195 n.m	[144 ft-lb]



Install the five capscrews that attach to the bottom of the oil pan adapter.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



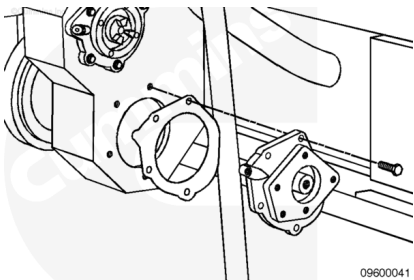
Install the hydraulic pump drive.

Use clean engine oil to lubricate the bushing in the front gear cover.

Install the gasket, drive or cover, washers, and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/56/56-001-023-tr.html>).
- Install the crankshaft adapter. Refer to Procedure 001-017 (</qs3/pubsys2/xml/en/procedures/56/56-001-017-tr.html>).
- Install the front engine support bracket. Refer to Procedure 016-002 (</qs3/pubsys2/xml/en/procedures/56/56-016-002-tr.html>).
- Install the vibration dampers. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html>).
- Install the accessory drive and alternator drive seal (if applicable). Refer to Procedure 001-003 (</qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html>).
- Install the alternator and accessory drive pulleys, if equipped. Refer to Procedure 009-004 (</qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html>).
- Install the LTA water pump. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Install the fan hub idler. Refer to Procedure 008-029 (</qs3/pubsys2/xml/en/procedures/56/56-008-029-tr.html>).
- Install the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html>).
- Install the alternator and bracket. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/56/56-013-001-tr.html>).
- Install the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html>).
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>).
- Install the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>).
- Install new full-flow oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).

tr.html).

- Connect the batteries. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/56/56-013-009.html).
- Connect the air starter to reduce the possibility of accidental starting. Refer to the OEM Service Manual.
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).
- Operate the engine and check for leaks.

with Electronically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Install the front crankshaft seal. Refer to Procedure 001-023 (/qs3/pubsys2/xml/en/procedures/56/56-001-023-tr.html).
- Install the crankshaft adapter. Refer to Procedure 001-017 (/qs3/pubsys2/xml/en/procedures/56/56-001-017-tr.html).
- Install the front engine support bracket. Refer to Procedure 016-002 (/qs3/pubsys2/xml/en/procedures/56/56-016-002-tr.html).
- Install the vibration dampers. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html).
- Install the accessory drive and alternator drive seal (if applicable). Refer to Procedure 001-003 (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html).
- Install the alternator and accessory drive pulleys, if equipped. Refer to Procedure 009-004

(/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html).

- Install the fan hub idler. Refer to Procedure 008-029 (/qs3/pubsys2/xml/en/procedures/56/56-008-029-tr.html).
- Install the fan hub. Refer to Procedure 008-036 (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html).
- Install the alternator and bracket. Refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/56/56-013-001-tr.html).
- Install the alternator belt. Refer to Procedure 013-005 (/qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html).
- Install the fan belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html).
- Install the fan and fan spacer(s). Refer to Procedure 008-040 (/qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html).
- Connect the batteries. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/56/56-013-009.html).
- Connect the air starter to reduce the possibility of accidental starting. Refer to the OEM Service Manual.
- Operate the engine and check for leaks.